

Our second midterm exam will be held on Monday, April 7th in class. You will have the entire class period to work the exam. Some of you may finish early, and if so, you are free to leave once your exam has been submitted.

If you have accommodations and need to schedule a time to take the test in the testing center, please do so.

I do not normally allow for rescheduling a test except in extenuating circumstances. Please communicate if you need to discuss rescheduling. Appropriate reasons to request an alternative test date include illness, participation in a Hood sporting event that conflicts with our exam date, etc. **No-shows on the day of the exam will earn a score of zero.** Students who arrive late will not be allowed extra time at the end.

This exam will primarily focus on the material we have covered since the first midterm exam, though it will necessarily overlap with the content from the first midterm exam: as you have been seeing, that material on data basics, experimental design, describing distributions is still coming up over and over again as we have been studying newer material.

You may use one double-sided 8.5 x 11in sheet of notes that should be handwritten by you yourself, not typed. (If you have a documented accommodation that allows for typed notes, of course, you may use your accommodation.) You may use a calculator, but not your phone. If you need to borrow a calculator, please ask. No collaboration is allowed as this exam should assess your personal understanding so that I can properly award you credit for your learning.

To study for this test, I strongly recommend that you pull out your old homework assignments, labs, and quizzes, hide the answers from yourself and re-do the work, checking your answers against the solutions that are posted on Blackboard. The exercise sections in our textbook are another good source of problems. Remember that answers to odd-numbered exercises are at the “back” of the book.

Things to know and be able to do

Data basics and study design from Exam 1 material

- Be able to identify cases, variables (numerical or categorical) in a study.
- Be able to identify the population, sample, parameters, statistics, explanatory (or predictor) variables, response variables in a study.
- Be able to comment on whether a particular study lends itself to generalization and / or to drawing causal conclusions.

Describing, summarizing, visualizing single variables from Exam 1 material

- Numerical variables
 - Given a visualization of the distribution of a single numerical variable, be able to write a description of the distribution, first writing a who/what/where/when sentence, and then continuing the paragraph addressing center, shape, spread, outliers, and any other interesting features. All such descriptions should be in context, and should be supported with numbers from the graph or from computed statistics.
 - Be able to use appropriate measures of center (mean or median) and spread (standard deviation or quantiles (middle 50% and middle 95%)). Remember that you should use different combinations of these depending on whether your distribution is symmetric or skewed. Mean + standard deviation are good for symmetrical distributions. Median + quantiles are good for skewed distributions.
 - Be able to say which sort of graph would be appropriate for a particular visualization.
- Categorical variables
 - Given a visualization of a categorical variable, or a pair of categorical variables, be able to write a paragraph highlighting key features of the distribution, speaking in context, and supporting your points with numbers drawn from the graph.
 - Given a contingency table, be able to compute row and column proportions to back up claims you make about a study.
 - Be able to comment on whether there is an association between two categorical variables, given a visualization or a contingency table for these variables.

Linear regression - numerical explanatory and numerical response

- Given a research question having a numerical explanatory (predictor) and numerical response, know that a scatterplot is the appropriate visualization.
- Be able to describe the relationship between two numerical variables, given a scatterplot depicting the relationship. Descriptions should address shape, strength, and direction of the association, should use the “General trends, Examples, Exceptions” (GEE) model for description, and should be in context. Examples illustrating the strength of the association should provide examples of the variability of the response variable for given values of the explanatory variable, as we have done in class and in homework solutions.
- Be able to use the R model summary output to write the regression equation.
- Be able to write a sentence of interpretation (in context) for the slope of a regression line. I’m looking for a sentence of very specific format here, as we practiced in class and on homework.

- Be able to write a sentence of interpretation (in context) for the intercept of a regression line. I'm looking for a sentence of very specific format here, as we practiced in class and on homework.
- Be able to use the equation of a regression line to find a predicted value of the explanatory variable. Care must be taken to include units, for full credit.
- Be able to calculate the residual of a data point, showing all work. Care must be taken to include units for full credit.
- Just by looking at a data point on a scatterplot where the relationship has been modeled using a line, be able to say whether the residual will be positive or negative.
- Be able to estimate a value for r , the correlation coefficient, from a scatterplot

Linear regression – categorical predictors and multiple predictors

- Given regression model summary output involving a single categorical predictor and a numerical response, be able to write the regression equation.
- Given regression model summary output involving a single categorical predictor and a numerical response, be able to write sentences of interpretation for the coefficients of the model, in context, and in the style we have practiced in class and on homework.
- Given regression model summary output involving a categorical predictor and a numerical predictor for a numerical response variable, be able to write the regression equation.
- Given regression model summary output involving a categorical predictor and a numerical predictor for a numerical response variable, be able to write sentences of interpretation for the coefficients of the model, in context, and in the style we have practiced in class and on homework.
- Be able to interpret the meaning of an R^2 value for a model, as we have practiced in class.

Learning from experiments: comparing effects of treatments on groups

- Given the description of a study with control and treatment groups, be able to analyze and answer the question of whether it seems plausible that the observed difference between the treatment groups is due to the natural variability associated with the random assignment of subjects to treatment groups. (E.g. Sex Discrimination study, Weight Loss study). More specifically, you should be able to:
 - Be able to write a null and alternate hypothesis, both in words and using correct notation.

- For a given dotplot for such a simulation, be able to tell exactly what each dot represents.
- Be able to write a paragraph describing the dotplot, in context.
- Be able to estimate and report a one-sided or two-sided p-value.
- Be able to write a sentence telling how the p-value influences your answer to the question of compatibility of the observed difference with the null model.

Learning from experiments: assessment of the null model in a study involving one proportion

- Given the description of a study in which the statistic under consideration is a single proportion, be able to analyze and answer the question of whether the null model of no effect provides a plausible explanation for the observed proportion. (E.g. Helper/Hinderer study, ESP study). More specifically,
 - Be able to write a null and alternate hypothesis, both in words and using the correct notation.
 - For a given dotplot for such a simulation, be able to tell exactly what each dotplot represents.
 - Be able to write a paragraph describing the dotplot, in context.
 - Be able to compute and report a one-sided or two-sided p-value.
 - Be able to write a sentence telling how the p-value influences your answer to the question of compatibility of the observed proportion with the null model.

Sampling distributions

We will save sampling distributions for examination on the final exam.